

ALAN ZHOU

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EDUCATION

Johns Hopkins University
PhD in Cognitive Science

Baltimore, MD
Aug. 2022 - July 2027

University of California, Berkeley
B.A. in Computer Science and Cognitive Science

Berkeley, CA
Aug. 2017 - Dec. 2021

PUBLICATIONS

* denotes equal contribution

In Press

- (2026) Alan Zhou*, Miloš Stanojević*, and John Hale. Syntactic Belief Update as the Driver of Garden Path Processing Difficulty. To appear in *Proceedings of EMNLP 2026*. [arXiv](#) 

Peer-reviewed Journals and Conference Proceedings

- (2026) Jane Li* and Alan Zhou*. (R)NNs too expressive? No relationship between neural architectures' theoretical expressivity and their empirical performance. In *Linguistics Vanguard*. [PDF](#) 
- (2024) Alan Zhou and Colin Wilson. Modeling morphosyntactic agreement as neural search: a case study of Hindi-Urdu. In *Proceedings of the Society for Computation in Linguistics 2024*, pages 227–239. [PDF](#) 
- (2023) Gašper Beguš*, Alan Zhou*, Peter Wu, and Gopala K Anumanchipalli. Articulation GAN: Unsupervised modeling of articulatory learning. In *ICASSP 2023 IEEE International Conference on Acoustics, Speech and Signal Processing*. [PDF](#)  [Video](#) 
- (2023) Gašper Beguš, Alan Zhou, and Christina Zhao. Encoding of speech in convolutional layers and the brain stem based on language experience. *Scientific Reports*. [PDF](#) 
- (2022) Gašper Beguš and Alan Zhou. Interpreting intermediate convolutional layers of generative CNNs trained on waveforms. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 30. [PDF](#) 
- (2022) Gasper Begus and Alan Zhou. Modeling speech recognition and synthesis simultaneously: Encoding and decoding lexical and sublexical semantic information into speech with no direct access to speech data. In *Proc. Interspeech 2022*. [PDF](#) 
- (2022) Gašper Beguš and Alan Zhou. Interpreting intermediate convolutional layers in unsupervised acoustic word classification. In *ICASSP 2022 IEEE International Conference on Acoustics, Speech and Signal Processing*. [PDF](#) 

PRESENTATIONS

- (2025) Jane Li and Alan Zhou. CNNs that robustly compute vowel harmony do not explicitly represent tiers. Poster at *Society for Computation in Linguistics*. [Abstract](#) 
- (2025) Gašper Beguš, Thomas Lu, Alan Zhou, and Gopala Anumanchipalli. CiwaGAN: Informative imitation with articulatory learning. Poster at *Linguistic Society of America Annual Meeting*.
- (2025) Alan Zhou, Bruno Ferenc Segedin, Jingyi Chen, Sneha Ray Barman, and Gašper Beguš. Deep Language Learning: Modeling language from raw speech. Tutorial at *Linguistic Society of America Annual Meeting*. [Slides and Materials](#) 

AWARDS AND HONORS

Dissertation Prize Fellowship Spring 2027
JHU Office of the Dean of Arts and Sciences \$27,500 USD

Travel Grant May 2022
ICASSP 2022 (*declined*) \$500 USD

TEACHING

At Johns Hopkins

• **Foundations of Cognitive Science**, *Teaching Assistant* Fall 2025

• **Foundations of Neural Network Theory**, *Teaching Assistant* Spring 2024

- Led weekly recitation sessions offering help for homework and course content
- Created and presented a set of three lectures introducing students to basic Python programming and constructing neural network models in PyTorch
- Created a corresponding set of Python problem sets for each lecture.

• **Bayesian Inference**, *Teaching Assistant* Fall 2023, 2024

- Presented a lecture on the Rational Speech Acts model of pragmatic inference (2024 offering)

• **Neuroscience: Cognitive**, *Teaching Assistant* Spring 2023

At UC Berkeley

• **Deep Learning and Phonology**, *Guest Lecturer* Fall 2021

- Presented a guest lecture on high-performance computing to linguistics graduate students

• **Data Structures**, *Academic Intern* Spring 2018

- Helped students in lab sections and office hours

SERVICE

Department Lead, Early Career Colloquium Committee Fall 2024 to Present

- Led organization of a colloquium talk series meant to highlight the work of early postdocs and late PhD students across cognitive science
- Successfully secured funding from the Dean's office and coordinated visits

Member, General Colloquium Committee Fall 2025 to Present

- Helped coordinate student hosts for general colloquium visits

SKILLS

Programming Languages:	Python, Java, C, MATLAB, R, Lua, SQL
Tools/Technologies:	PyTorch, Tensorflow, Keras, Slurm, matplotlib Jupyter, Git, Gradle/Maven
Natural Languages:	English (fluent), Mandarin (conversational)